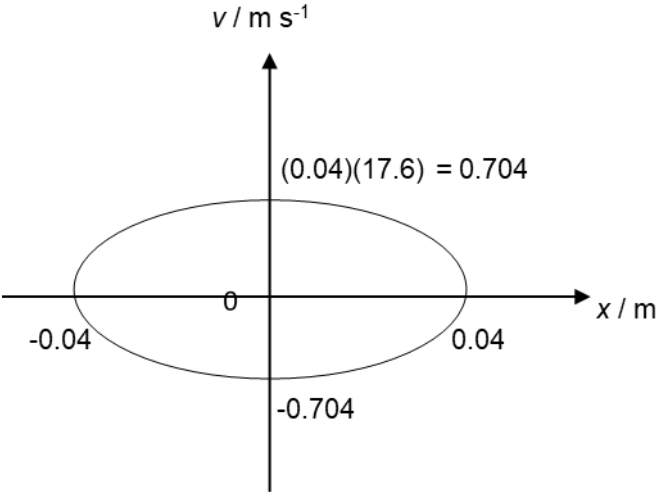


2(a)(i)	$\omega = 2\pi/T$ (where $T = 10/28$ s) $= 17.6 \text{ rad s}^{-1}$ (accept if T is read off graph)	B1
aii	 correct shape x_0 and v_0 labelled	M1 A1
(a)(iii)	At $t = 0$, x = its max positive value $\rightarrow$ eqn of motion is of the form $x = x_0 \cos \omega t$ rather than $x = x_0 \sin \omega t$ From Fig 3.2, $x_0 = 0.04$ m. $\rightarrow x = 0.04 \cos (17.6 t)$ (accept if T is read off graph)	B1 B1
a)(iv)	Graph showing <u>amplitude decreasing</u> with time (must be consistent for all that peaks both above and below the t axis) Period remains unchanged or slightly larger.	B1 B1
(b)(i)	$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.20}{6.0}} = 1.15 \approx 1.2 \text{ s}$	B1
(b)(ii)	Max speed $v_{\max} = \omega x_0$, where $x_0 = 0.05$ m, $\omega = \frac{2\pi}{T}$ $= 0.27 \text{ m s}^{-1}$	C1 A1
(b)(iii)	Total energy $= \frac{1}{2} m v_{\max}^2 = \frac{1}{2} (0.200)(0.273)^2$ $= 7.5 \times 10^{-3} \text{ J}$	C1 A1

2(e)(i)	The electric force per unit positive charge placed at that point	D1
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